

PROJECT

CITY CARE HOSPITAL

Real-Time Hospital Appointment & Queue Management System

Domain: Distributed Healthcare Systems, Real-Time Web Engineering, Enterprise Java

Architecture: Model-View-Controller (MVC) Pattern & Asynchronous Polling Architecture

Platforms: Apache Tomcat 10.1 EE & MySQL 8.0 Relational Database (Cloud & Container Ready)

1. PROJECT ABSTRACT & DESCRIPTION

Abstract :

The Real-Time Hospital Appointment and Queue Management System (City Care Hospital) is an enterprise-grade outpatient department (OPD) queue coordination and token management platform. The system models a synchronized clinical workflow connecting outpatient visitors, attending medical specialists, and hospital administrative staff. Patients can browse specialized clinical departments (General Medicine, Cardiology, Orthopedics, Pediatrics), inspect doctor availability schedules, and reserve numbered consultation tokens without physical waiting line congestion. When patients arrive, attending physicians manage their live consultation queue via an interactive clinical console, calling tokens sequentially and advancing status flags (Waiting → Called → In Consultation → Completed / Absent). A lightweight asynchronous polling engine (AJAX Fetch API at 3-second intervals) keeps patient status displays in real-time synchronization with doctor actions without full-page reloads. Additionally, the system features an enterprise XML configuration subsystem with Java DOM Parser (DocumentBuilderFactory), dual DTD/XSD schema verification, and real-time XPath query evaluation.

Motivation & Problem Statement :

- **Outpatient Hall Congestion & Extended Wait Times:** Traditional OPD queues force vulnerable patients into overcrowded waiting halls, increasing cross-infection contagion risks, patient anxiety, and operational friction.
- **Lack of Real-Time Queue Transparency:** Patients historically lack visibility into how many individuals precede them or when their turn is imminent, preventing timely arrival and disrupting doctor schedules.
- **Scheduling Clashes & Token Double-Booking:** Uncoordinated manual token dispatch often leads to token number collisions, overbooked clinic hours, and high doctor burnout without automated guardrails.
- **Comprehensive Enterprise Web Standards Implementation:** Serves as an end-to-end realization of modern Web Technology standards—combining classical Java Enterprise architecture (Jakarta Servlet 6.0, JSP 3.1, JDBC, XML DOM, DTD/XSD validation, XPath) with dynamic client-side DOM mutation and asynchronous Fetch APIs.

2. SOFTWARE SPECIFICATIONS

Tier / Layer	Technology	Version
Server Runtime (Java)	Apache Tomcat	10.1.x
Language (Backend)	Java SE (JDK)	21 LTS / 17 LTS
Java Standards	Jakarta Servlet & JSP	6.0 / 3.1
Database Tier	MySQL Relational Database	8.0.x
Asynchronous Engine	AJAX (JavaScript Fetch API)	ES6+ Modern Standard
Data Interchange	XML, DTD, XSD & Java DOM Parser	W3C DOM / JAXP
Query & Path Evaluation	XPath API (javax.xml.xpath)	W3C XPath 1.0/2.0
Security & Cryptography	PBKDF2 with HMAC-SHA256	RFC 2898 Standard
State & Personalization	HTTP Session & Persistent Cookies	RFC 6265 Standard
Build Automation	Apache Maven	3.9.x
Client Presentation	HTML5, CSS3 Modern Tokens, Vanilla JS	W3C Standards

3. SYSTEM ARCHITECTURE & DESIGN PATTERNS

Client Web Browser (HTML5 / CSS3 / JavaScript Fetch API) HTTP Requests (GET/POST) / Asynchronous Polling (3s Interval) Authentication Filter (AuthFilter)
+-----+-----+-----+ Patient
Servlets Doctor Servlets Admin Servlets - BookTokenServlet - CallNextPatientServlet -
DepartmentServlet - AvailableTokensServlet - StartConsultationServlet - DoctorServlet -
QueueStatusServlet - CompleteConsultationServlet - RoomServlet - CheckInServlet - MarkAbsentServlet -
PatientServlet - CancelTokenServlet - DoctorQueueServlet - XmlConfigServlet
+-----+-----+-----+ Business Service
Layer & XML Subsystem - HospitalConfigParser (DOM DocumentBuilder) - HospitalXPathService (XPath
Evaluator) - XMLValidator (DTD & XSD Validators) - PasswordUtil (PBKDF2 HMAC-SHA256) Data Access
Objects (DAO) [UserDAO] [DoctorDAO] [QueueDAO] [DepartmentDAO] [RoomDAO] JDBC Connection Pool
(DBConnection) MySQL Relational Database (hospital_queue_db)

Design Patterns Implemented:

1. **Model-View-Controller (MVC):** Decouples presentation (JSP / modern CSS / DOM rendering), request routing and controller handling (Jakarta Servlets), and persistent domain entity models (Java Beans).
2. **Data Access Object (DAO) Pattern:** Encapsulates all raw SQL statements and transactional operations into dedicated classes (QueueDAO, DoctorDAO, UserDAO, DepartmentDAO), ensuring database independence and clean separation of concerns.
3. **Singleton Connection Pattern:** DBConnection.java centralizes database connection retrieval, managing socket life-cycles, credentials, and connection health across concurrent requests.
4. **Observer & Polling Pattern:** Lightweight client-side asynchronous Fetch loops poll the server every 3 seconds to observe queue state transitions, dynamically updating DOM nodes without page reloads.
5. **Intercepting Filter Pattern:** AuthFilter.java intercepts every incoming HTTP request to enforce strict Role-Based Access Control (RBAC), preventing privilege escalation across Patient, Doctor, and Admin zones.

4. WORKFLOW & SYSTEM LIFECYCLE

1. Authentication & Session Initialization:

- Patients, Doctors, and Administrators log in via a unified secure entry point at /auth/login.jsp.
- Credentials are authenticated against PBKDF2 with HMAC-SHA256 salted password hashes stored in MySQL.
- A secure HttpSession is established, persisting user ID, full name, and granted security role.
- Optional 30-day 'Remember Me' and clinic preference cookies are stored securely in the client browser.

2. Clinic Selection & Online Token Booking:

- Patient selects a specialty department (e.g., General Medicine, Cardiology, Orthopedics, Pediatrics).
- An asynchronous AJAX call retrieves available specialists, room assignments, and consultation dates.
- The client renders an interactive 30-token slot grid showing available, booked, and current queue tokens.
- Upon selection, an atomic database transaction reserves the token, preventing race condition double-bookings.

3. Real-Time Queue Synchronization & Live Status Tracking:

- Patient accesses the Live Queue or Dashboard, which initiates background 3-second polling.
- The system continuously calculates and returns: Current Calling Token, People Ahead, and Estimated Wait Time.
- Threshold alerts trigger instantly: When 0 patients remain, the banner signals '**YOU ARE NEXT**'.
- When the token is actively called, the display updates to '**YOUR TOKEN IS CALLED. PROCEED TO ROOM [X]**'.

4. Doctor Clinical Console & Queue Progression:

- The attending physician views their live roster of waiting patients ordered sequentially by token number.
- Doctor clicks '**Call Next Patient**', updating queue status from WAITING to CALLED.
- Doctor marks '**Start Consultation**' (advances to IN_CONSULTATION) and '**Complete**' (advances to COMPLETED).
- Unresponsive patients can be flagged as ABSENT, automatically advancing the waiting line.

5. Administrative Oversight & XML Configuration:

- Administrator manages hospital metadata, department departments, rooms, and doctor assignments.
- The XML console parses hospital-config.xml using Java DOM Parser (DocumentBuilder).
- Live DTD and XSD validation controls verify document syntax, schema constraints, and structural integrity.
- An XPath evaluation console executes real-time node queries across departments and operational parameters.

5. CORE FEATURES BREAKDOWN

Feature 1: Asynchronous Real-Time Queue Synchronization

- Client-side `setInterval()` poller queries `/patient/queue-status` every 3,000 milliseconds.
- Updates Current Token, Number of Patients Ahead, and Estimated Minutes remaining with zero full-page reloads.
- Dynamic badge and alert mutations keep waiting patients informed in real time.

Feature 2: Concurrency-Safe Online Token Reservation

- Interactive digital token grid rendering 30 daily consultation slots per physician.
- Backend unique composite constraint `UNIQUE (doctor_id, queue_date, token_number)` guarantees no duplicate allocations.
- Instant booking confirmation with room routing, consultation date, and persistent token summary.

Feature 3: Role-Based Access Control (RBAC) & Multi-Role Architecture

- Three distinct privilege levels: **PATIENT**, **DOCTOR**, and **ADMIN**.
- `AuthFilter` intercepts and protects restricted URI paths (`/patient/*`, `/doctor/*`, `/admin/*`).
- Automatic redirect to login with session expiry detection and unauthorized access blocking.

Feature 4: Physician Clinical Console & Queue Orchestration

- Dedicated doctor workstation displaying waiting patient list, token progression, and daily consultation totals.
- Action triggers: *Call Next Patient*, *Start Consultation*, *Complete Consultation*, and *Mark Absent*.
- Doctor availability toggle (AVAILABLE vs. OFFLINE) reflects immediately on public rosters.

Feature 5: Clinical Department Directory & Specialist Routing

- Multi-specialty support: General Medicine (Room 101), Cardiology (Room 202), Orthopedics (Room 305), Pediatrics (Room 401).
- Full CRUD management of departments, physical rooms, and doctor clinic schedules.

Feature 6: Dynamic XML Hospital Configuration (Java DOM Parser)

- Centralized hospital configuration maintained in standard `hospital-config.xml`.
- Backend parsing via Java JAXP `DocumentBuilderFactory` and `DocumentBuilder`.
- Extracts hospital metadata, average consultation durations, capacity caps, and department definitions.

Feature 7: DTD & XSD Dual Schema Validation Engine

- Structural validation against Document Type Definition (hospital-config.dtd).
- Strict schema validation against XML Schema Definition (hospital-config.xsd) for types and ranges.
- Live admin console executing asynchronous validation requests and reporting validation status.

Feature 8: Real-Time XPath Query Evaluator

- Executes real-time XPath expressions using Java javax.xml.xpath.XPath.
- Supports parameterized queries (e.g., //department[name='Cardiology'], //department[averageTime > 5]).
- Renders query result nodes directly into formatted administrative data tables.

Feature 9: Patient Preference Persistence & Session Management

- Custom CookieUtil handles client preferences (Preferred Department, Preferred Language, Display Mode).
- Secure session tracking with CSRF safeguards, session fixation protection, and PBKDF2 password cryptography.

Feature 10: Executive Administration & Hospital Monitoring

- Executive dashboard displaying today's patients, online bookings, walk-in tokens, and active clinics.
- Global queue monitor tracking real-time queue states across all hospital consultation rooms simultaneously.

6. CONCLUSION & FUTURE ENHANCEMENTS

The Real-Time Hospital Appointment and Queue Management System successfully bridges classical enterprise Java architectural patterns (Jakarta Servlets, JSP, JDBC, XML DOM, DTD/XSD, XPath) with modern responsive user interfaces and asynchronous polling workflows. By replacing manual waiting hall crowding with transparent digital tokens and sub-second queue tracking, the platform dramatically reduces patient anxiety, optimizes doctor utilization, and provides hospital leadership with continuous operational visibility.

Future Enhancements :

- **SMS & WhatsApp Notification Gateway:** Dispatching automated SMS alerts when a patient is 2 tokens away, allowing patients to arrive just in time from outside the hospital premises.
- **Automated Kiosk & QR Code Check-in:** Allowing patients with online bookings to scan a dynamic QR code at the hospital lobby kiosk to confirm arrival and activate their queue slot.
- **Machine Learning Wait Time Prediction:** Leveraging historical consultation durations, patient age, and diagnostic categories to dynamically refine estimated waiting times beyond fixed averages.
- **Integrated Telemedicine & E-Prescription:** Enabling remote audio/video consultations for follow-up patients with automated PDF prescription generation.

7. SYSTEM USER INTERFACE & WORKFLOW WALKTHROUGH

The following representations illustrate the core interactive modules deployed across the application:

Figure 1: Hospital Landing Portal & Patient Entry (index.jsp)

A modern hospital homepage featuring hospital branding, quick action buttons (Book Token, Live Queue, Doctor Directory), 3-step patient journey guide, active departments directory with direct booking links, and 24/7 emergency support contact information.

+ City Care Hospital • Home Book Token Live Queue Login Register
Your Health, On Time. Without the Waiting. Reserve your consultation token online, track your turn live from your phone, and arrive exactly when the doctor is ready.
[+ Book Token] [• View Live Queue] [Find a Doctor]
Clinical Specialties: • General Medicine (Room 101) – Dr. Kumar (Available) • Cardiology (Room 202) – Dr. Priya (Available) • Orthopedics (Room 305) – Dr. Arun (Available) • Pediatrics (Room 401) – Child Healthcare Specialists

Figure 2: Concurrency-Safe Token Booking Interface (book-token.jsp)

Interactive token booking interface where patients select department, doctor, and date. A dynamic 30-token slot grid displays booked vs available slots, ensuring real-time slot selection with zero duplicate collisions.

Book Consultation Token – Dr. Kumar (General Medicine)
Selected Date: Today Room: 101 Est. Time per Patient: 5 mins
[01 Booked] [02 Booked] ... [20 CALLED] [21 Waiting] [22 Waiting] [25 SELECTED - YOUR TOKEN] [26 Available] [27 Available] ... [30 Available]
Available Slots: 5 of 30 [Confirm & Book Token #25]

7. SYSTEM USER INTERFACE & WORKFLOW WALKTHROUGH (CONT.)

Figure 3: Real-Time Patient Live Queue & Status Monitor (dashboard.jsp)

Patient dashboard polling every 3 seconds via AJAX. Displays active token number, doctor room, calling status, countdown of patients ahead, and estimated wait minutes with instant alert transitions.

Welcome, Arun • Live Queue Consultation Status (Auto-Syncing Active)
[ALERT] Please wait. 4 patients are ahead of you. (Status: WAITING)
Your Token: #25 Doctor: Dr. Kumar Room: 101 Current Calling Token: #20 People Ahead: 4 Est. Wait Time: 20 mins
Dynamic Alert States: 1 Ahead → '1 patient ahead' 0 Ahead → 'YOU ARE NEXT' Called → 'YOUR TOKEN IS CALLED. PROCEED TO ROOM 101'

Figure 4: Doctor Clinical Console & Admin XML Configuration Hub

Doctor console for advancing queue states, alongside the Admin XML Console featuring DOM parsing, DTD/XSD validation buttons, and real-time XPath evaluation against hospital-config.xml.

Doctor Console (Dr. Kumar • Room 101)
Currently In Consultation: Token #20 (Completed) Next in Line: Token #21 (Waiting) [• Call Next Patient] [Start Consultation] [Finish Consultation] [Mark Absent]
Admin XML Configuration & Verification Console (xml-config.jsp)
Hospital Master Configuration: hospital-config.xml [Active & Synchronized] [Run DTD Validation] → VALID: hospital-config.xml conforms to hospital-config.dtd [Run XSD Validation] → VALID: Schema conforms to hospital-config.xsd - XPath Query: //department[averageTime > 5] → [Execute XPath] Results: Cardiology (10 mins), Orthopedics (8 mins), Pediatrics (6 mins)